

Pana ERP v3.0 - End-to-End UI/UX Testing Guide

Version: 1.0.0
Created: 2026-01-27
Purpose: Complete workflow testing from Lead to Payment
Target Audience: Developers, QA, Client Demo
Estimated Time: 2-3 hours for full walkthrough

Overview

This document provides a step-by-step guide to test the complete Pana ERP v3.0 workflow, simulating a real printing job from initial customer inquiry to final payment collection.

Scenario Used: "Pana Promotion receives a phone call from a new client requesting 500 business cards. The job goes through quotation, approval, manufacturing, delivery, and invoicing."

Pre-Test Checklist

Before starting the workflow test, ensure:

- ☐ Development server is running (`npm run dev`)
- ☐ ERPNext backend is accessible
- ☐ You can login to the dashboard
- ☐ The following master data exists:
 - At least 1 Company
 - At least 1 Customer Group
 - At least 1 Territory
 - At least 1 Item Group (Raw Materials, Finished Goods)
 - At least 1 Warehouse (Raw Materials, Finished Goods)

Demo Data Setup (If Starting Fresh)

Step 0.1: Create Required Settings

Setting	Path	Data to Create
Customer Group	<code>/crm/settings/customer-group</code>	"Retail", "Wholesale"
Territory	<code>/crm/settings/territory</code>	"Addis Ababa"
Lead Source	<code>/crm/settings/lead-source</code>	"Walk-in", "Phone", "Referral"
Item Group	(ERPNext setup or API)	"Raw Materials", "Finished Goods", "Services"

Step 0.2: Create Raw Material Items

Navigate to </stock/item/new>

Item 1: A4 Card Stock

```
item_code: RM-CARD-A4
item_name: A4 Card Stock 300gsm
item_group: Raw Materials
stock_uom: Sheet
is_stock_item: ✓ (checked)
is_purchase_item: ✓ (checked)
is_sales_item: ✗ (unchecked)
standard_rate: 5 (ETB per sheet)
```

Item 2: Business Card Template

```
item_code: FG-BCARD-STD
item_name: Standard Business Cards
item_group: Finished Goods
stock_uom: Nos
is_stock_item: ✓ (checked)
is_purchase_item: ✗ (unchecked)
is_sales_item: ✓ (checked)
standard_rate: 0.50 (ETB per card)
```

Step 0.3: Create Warehouses

Navigate to </stock/warehouse/new>

```
Warehouse 1: Raw Material Store
Warehouse 2: Finished Goods Store
Warehouse 3: WIP - Production
```

Step 0.4: Create Initial Stock

Navigate to </stock/stock-entry/new>

```
Stock Entry Type: Material Receipt
Items:
- Item: RM-CARD-A4
  Qty: 100
  Target Warehouse: Raw Material Store
```

Test Scenario: Complete Print Job Lifecycle

Phase 1: CRM - Lead Capture & Conversion

Test 1.1: Create Lead

Navigation: Dashboard → CRM → Leads → New Lead

Action:

1. Click "New Lead" button
2. Fill in the form:

```
Lead Name: John Wondwosen  
Company Name: Alpha Trading PLC  
Mobile No: 0911234567  
Email: john@alphatrading.com  
Source: Phone  
Status: Open  
Notes: Interested in business cards for new office opening
```

3. Click "Create Lead"

Expected Result:

- ☐ Lead created successfully
- ☐ Toast notification appears
- ☐ Redirected to lead detail page
- ☐ Lead appears in list with "Open" status badge

Test 1.2: Update Lead Status

Navigation: CRM → Leads → Click on "John Wondwosen"

Action:

1. Click "Edit" button
2. Change Status to "Interested"
3. Save changes

Expected Result:

- ☐ Status badge updates to "Interested"
- ☐ Lead is now ready for conversion

Test 1.3: Convert Lead to Customer

Navigation: Lead detail page

Action:

1. Click "Convert to Customer" button

2. Review pre-filled customer form:

```
Customer Name: Alpha Trading PLC (from company_name)
Customer Type: Company
Customer Group: Retail
Territory: Addis Ababa
```

3. Click "Create Customer"

Expected Result:

- ☐ Customer created successfully
- ☐ Redirected to new Customer page
- ☐ Original Lead status changes to "Converted"

Test 1.4: Add Customer Address

Navigation: Customer detail page → Addresses & Contacts tab

Action:

1. Click "Add Address"
2. Fill in:

```
Address Title: Head Office
Address Type: Billing
Address Line 1: Bole Road, Building 45
City: Addis Ababa
Country: Ethiopia
Phone: 0911234567
```

3. Save address

Expected Result:

- ☐ Address created and linked to Customer
- ☐ Address appears in Customer's "Addresses" section
- ☐ Can be selected in Quotation later

Test 1.5: Add Customer Contact

Navigation: Customer detail page → Addresses & Contacts tab

Action:

1. Click "Add Contact"
2. Fill in:

```
First Name: John
Last Name: Wondwosen
Email: john@alphatrading.com
Mobile: 0911234567
Designation: Procurement Manager
Is Primary Contact: ✓
```

3. Save contact

Expected Result:

- ☐ Contact created and linked to Customer
 - ☐ Contact appears in Customer's "Contacts" section
-

Phase 2: Sales - Quotation & Order

Test 2.1: Create Quotation

Navigation: Sales → Quotations → New Quotation

Action:

1. Fill header:

```
Quotation To: Customer
Customer: Alpha Trading PLC
Transaction Date: [today]
Valid Till: [today + 15 days]
Company: Pana Promotion
```

2. Verify address/contact auto-populate

3. Add line item:

```
Item: FG-BCARD-STD
Description: 500 Business Cards, 300gsm, Full Color, Matt Lamination
            Design: Client to provide artwork
            Size: 9x5cm standard
Qty: 500
Rate: 1.50 ETB
```

4. (Optional) Select Tax Template

5. (Optional) Select Terms and Conditions

6. Click "Create"

Expected Result:

- ☐ Quotation created with status "Draft"
- ☐ Line item amount calculated ($500 \times 1.50 = 750$ ETB)
- ☐ Grand total displays correctly

Test 2.2: Submit Quotation

Navigation: Quotation detail page

Action:

1. Review all details
2. Click "Submit" button
3. Confirm submission

Expected Result:

- ☐ Status changes from "Draft" to "Open"
- ☐ "Submit" button becomes unavailable
- ☐ "Create Sales Order" button appears
- ☐ Document is now un-editable

Test 2.3: Create Sales Order from Quotation

Navigation: Quotation detail page

Action:

1. Click "Create Sales Order" button
2. Review pre-filled data from Quotation
3. Add required field:

Delivery Date: [today + 7 days]

4. **Artwork Check:** Verify the "Artwork Verified" checkbox requirement
5. Check the artwork verification box
6. Click "Create Sales Order"

Expected Result:

- ☐ Sales Order created
- ☐ All quotation data copied correctly
- ☐ Delivery date is set
- ☐ Quotation status updates to "Ordered"

Test 2.4: Submit Sales Order

Navigation: Sales Order detail page

Action:

1. Verify all details correct
2. Ensure artwork checkbox is checked
3. Click "Submit"

Expected Result:

- ☐ Status changes to "To Deliver and Bill"
 - ☐ Document locked for editing
 - ☐ "Create Delivery Note" action available
 - ☐ "Create Work Order" action available (if manufacturing)
-

Phase 3: Manufacturing (If Applicable)

Skip this phase if using pre-made stock items. Include if manufacturing is needed.

Test 3.1: Create BOM (Bill of Materials)

Navigation: Manufacturing → BOM → New BOM

Action:

1. Fill form:

Item: FG-BCARD-STD
Quantity: 100 (produces 100 cards)

2. Add raw materials:

Item: RM-CARD-A4
Qty: 2 (2 sheets make 100 cards)

3. Save and Submit BOM

Expected Result:

- ☐ BOM created showing cost calculation
- ☐ Raw material cost aggregated

Test 3.2: Create Work Order

Navigation: Manufacturing → Work Orders → New Work Order

OR from Sales Order: Click "Create Work Order"

Action:

1. Fill form:

```
Item: FG-BCARD-STD
BOM: [select the BOM created]
Qty to Manufacture: 500
Source Warehouse: Raw Material Store
Target Warehouse: Finished Goods Store
Sales Order: [link to SO if created from SO]
```

2. Submit Work Order

Expected Result:

- ☐ Work Order created
- ☐ Raw materials calculated (500 cards needs 10 sheets)
- ☐ Status: "Not Started"

Test 3.3: Execute Stock Entry (Manufacture)

Navigation: Work Order detail page → Create Stock Entry

OR: Stock → Stock Entry → New Stock Entry

Action:

1. Stock Entry Type: Manufacture
2. Link to Work Order
3. Verify items:

```
Source Items (Consumed):
- RM-CARD-A4, Qty: 10, From: Raw Material Store

Target Item (Produced):
- FG-BCARD-STD, Qty: 500, To: Finished Goods Store
```

4. Submit Stock Entry

Expected Result:

- ☐ Raw Material Store decreased by 10 sheets
- ☐ Finished Goods Store increased by 500 cards
- ☐ Work Order status updates to "Completed"

Phase 4: Delivery**Test 4.1: Create Delivery Note**

Navigation: Sales Order detail page → "Create Delivery Note"

OR: Stock → Delivery Notes → New

Action:

1. Pre-filled from Sales Order:

Customer: Alpha Trading PLC
Items from Sales Order auto-populated

2. Add delivery details (if logistics module active):

Driver: [select driver]
Vehicle: [select vehicle]

3. Verify items and quantities:

Item: FG-BCARD-STD
Qty: 500
Warehouse: Finished Goods Store

4. Save as Draft first

Expected Result:

- ☐ Delivery Note created in Draft
- ☐ Items linked to Sales Order

Test 4.2: Print Gate Pass

Navigation: Delivery Note detail page

Action:

1. Click "Print Gate Pass" button
2. Review print preview (should NOT show amounts - just items and quantities)
3. Print or save PDF

Expected Result:

- ☐ Gate pass displays without prices
- ☐ Shows: Customer, Items, Quantities, Driver info
- ☐ Suitable for security checkpoint

Test 4.3: Submit Delivery Note

Navigation: Delivery Note detail page

Action:

1. Click "Submit" button
2. Confirm

Expected Result:

- ☐ Status changes to "To Bill"
 - ☐ Stock in Finished Goods Store decreases by 500
 - ☐ Sales Order status updates to "To Bill" (if fully delivered)
 - ☐ "Create Invoice" action appears
-

Phase 5: Accounting

Test 5.1: Create Sales Invoice

Navigation: Delivery Note page → "Create Invoice"

OR: Accounting → Sales Invoices → New

Action:

1. Pre-filled from Delivery Note:

Customer: Alpha Trading PLC
Items: FG-BCARD-STD × 500 @ 1.50 ETB

2. Verify totals match
3. Add due date:

Due Date: [today + 30 days] (Net 30 terms)

4. Submit Invoice

Expected Result:

- ☐ Invoice created and submitted
- ☐ Status: "Unpaid" (or "Overdue" if past due)
- ☐ Grand Total: 750 ETB
- ☐ GL entries posted (Accounts Receivable debited)

Test 5.2: View Outstanding Invoices

Navigation: Accounting → Sales Invoices

Action:

1. Filter by "Unpaid"
2. Verify the Alpha Trading invoice appears

Expected Result:

- ☐ Invoice visible in list
- ☐ Outstanding amount shows correctly

Test 5.3: Create Payment Entry

Navigation: Accounting → Payment Entry → New

OR from Sales Invoice: "Create Payment"

Action:

1. Fill form:

```
Payment Type: Receive
Party Type: Customer
Party: Alpha Trading PLC
Paid Amount: 750 ETB
Mode of Payment: Cash (or Bank)
```

2. Click "Get Outstanding Invoices"
3. Verify invoice auto-selected
4. Submit Payment

Expected Result:

- ☐ Payment Entry created
- ☐ Invoice status changes to "Paid"
- ☐ GL entries posted (Cash/Bank debited, AR credited)

Test 5.4: Verify Full Cycle Completion

Navigation: Various pages

Checklist:

- ☐ Lead status: "Converted"
- ☐ Customer exists with Address/Contact
- ☐ Quotation status: "Ordered"
- ☐ Sales Order status: "Completed"
- ☐ Work Order status: "Completed" (if applicable)
- ☐ Delivery Note status: "Completed"
- ☐ Sales Invoice status: "Paid"
- ☐ Payment Entry exists and submitted

UI/UX Testing Notes

Cross-Module Navigation Tests

From	To	Method	Expected
Lead	Customer	"Convert" button	Pre-fill customer form
Customer	Quotation	"Create Quotation"	Pre-select customer
Quotation	Sales Order	"Create SO"	Copy all data
Sales Order	Delivery Note	"Create DN"	Copy items
Sales Order	Work Order	"Create WO"	Link production
Delivery Note	Invoice	"Create Invoice"	Copy items/amounts
Invoice	Payment	"Create Payment"	Pre-fill party, fetch invoice

Status Badge Color Verification

DocType	Status	Expected Color
Lead	Open	Blue
Lead	Interested	Green
Lead	Converted	Success/Green
Lead	Lost	Red
Quotation	Draft	Gray
Quotation	Open	Blue
Quotation	Ordered	Green
Sales Order	Draft	Gray
Sales Order	To Deliver	Blue
Sales Order	Completed	Green
Invoice	Unpaid	Yellow/Warning
Invoice	Overdue	Red
Invoice	Paid	Green

Theme Testing

For each page visited, toggle between light and dark mode:

- 1. Click theme toggle (sun/moon icon)
- 2. Verify:

- ☐ All text is readable
- ☐ Cards have proper background
- ☐ Borders are visible
- ☐ Status badges have correct colors
- ☐ Forms have proper input visibility

Responsive Design Testing

Test key pages at different viewport widths:

Width	Device	Test Priority
375px	Mobile	Forms usable, cards stack
768px	Tablet	Side-by-side where appropriate
1280px	Desktop	Full layout

Error Scenario Testing

Test: Create Invalid Quotation

Action: Try to create quotation without items

Expected:

- ☐ Validation error: "At least one item required"
- ☐ Form does not submit

Test: Delete Submitted Document

Action: Try to delete a submitted Sales Order

Expected:

- ☐ Error: "Cannot delete submitted document"
- ☐ Document remains

Test: Negative Stock

Action: Try to deliver more than available stock

Expected:

- ☐ Error: "Insufficient stock for item X in warehouse Y"
- ☐ Delivery Note not submitted

Test: Credit Limit Warning

Action: Create invoice that exceeds customer credit limit (if configured)

Expected:

- ☐ Warning displayed
- ☐ User can still proceed (warning, not block)

Test: Unbalanced Journal Entry

Action: Create journal entry with unequal debits/credits

Expected:

- ☐ Error: "Total Debit must equal Total Credit"
- ☐ Cannot submit

Performance Benchmarks

Record load times for key pages:

Page	Target Load Time	Notes
Dashboard	< 2s	
Customer List (100 items)	< 3s	
Item List (500 items)	< 3s	
Quotation Create	< 1s	Form load
Sales Invoice Detail	< 2s	

Post-Test Report Template

```
## Pana ERP v3.0 E2E Test Report

**Date:** [YYYY-MM-DD]
**Tester:** [Name]
**Environment:** [Dev/Staging/Production]
**Browser:** [Chrome/Firefox/Safari] v[X]

### Summary
- Total Tests: X
- Passed: X
- Failed: X
- Blocked: X

### Failed Tests
| Test ID | Description | Expected | Actual | Screenshot |
|-----|-----|-----|-----|-----|
| 2.1 | Create Quotation | Success | Error on save | link |

### Bugs Found
| Bug ID | Severity | Description | Steps to Reproduce |
|-----|-----|-----|-----|
```

```
| BUG-001 | High | ... | 1. 2. 3. |

### UX Issues
| Issue | Page | Suggestion |
|-----|-----|-----|
| Button hard to find | SO Detail | Move to top action bar |

### Notes
[Any additional observations]
```

Client Demo: Business Gains & Value Proposition

Purpose: During the demo, communicate these business benefits to help the client understand the value of Pana ERP beyond the features.

💰 Projected Business Gains (Addis Ababa Context)

Use these talking points during the demo to connect features to real business value.

1. Time Savings

Current Pain Point	ERP Solution	Estimated Savings
Manual quotation writing in Word/Excel	Auto-calculate, one-click quotes	2-3 hours/day saved
Searching paper files for customer history	Instant customer hub with all records	30+ min/day saved
Calling warehouse to check stock	Real-time stock visibility	1 hour/day saved
Manual invoice preparation	One-click invoice from delivery	1-2 hours/day saved

Demo Script: "Ato [Client], right now when a customer calls asking about a past order, how long does it take to find that information? With this system, it's 3 seconds - just search the customer name."

2. Revenue Protection (Preventing Leakage)

Risk	ERP Prevention	Impact
Forgetting to invoice a delivery	Delivery Note automatically triggers "To Bill" status	Prevents unbilled revenue
Giving incorrect quotes from memory	Standard rate cards, linked items	Prevents underpricing by 5-10%
Lost leads due to no follow-up	Lead status tracking, visible pipeline	Recover 10-20% lost opportunities
Delivering without payment on credit customers	Credit limit warnings	Reduce bad debt exposure

Demo Script: "Last month, how many deliveries happened where the invoice was delayed or forgotten? Each one of those is money sitting outside your bank account. This system makes it impossible to forget."

3. Operational Visibility (Manager Benefits)

Question	Before ERP	With ERP
"How much did we sell this week?"	Manual calculation from receipts	Dashboard shows instantly
"Do we have paper stock for this job?"	Walk to warehouse, count	Real-time stock levels on screen
"Which jobs are overdue?"	Check with each department	Overdue Sales Orders highlighted in red
"Who owes us money?"	Review paper ledger	Outstanding Invoices report, one click

Demo Script: "Imagine starting every morning knowing exactly which jobs are overdue, which customers owe money, and what stock you're running low on - without asking anyone."

4. Cost Reduction

Expense	ERP Impact	Estimated Savings
Paper forms (quotations, invoices, DN)	Digital documents, print on demand	ETB 5,000-10,000/month
Staff overtime for end-of-month accounting	Automated ledger, no month-end scramble	10-20 hours/month
Expedited purchases due to stock-out	Low stock alerts, planned purchasing	5-10% on rush orders
Wasted materials from duplicate orders	Clear Work Order tracking	Reduce waste 5-15%

Demo Script: "Every time you run out of paper and have to buy emergency stock at a higher price from wherever is closest - that's money lost. This system tells you 3 days before you run out."

5. Professionalism & Customer Trust

Old Way	New Way	Customer Perception
Handwritten quotation on letterhead	Professional printed quotation with company logo	"This is a serious company"
Verbal delivery confirmation	Signed Delivery Note with clear items	"They're organized"
Delayed/inconsistent invoicing	Same-day invoice after delivery	"They're professional"
No record of past orders	"Let me pull up your history..."	"They remember me"

Demo Script: "When a customer from Bole calls and you instantly know their company name, their last 5 orders, and their preferred paper stock - how does that make your business look? World-class."

ROI Calculation Template

Use this during the demo to calculate rough ROI for the specific client:

MONTHLY SAVINGS ESTIMATE

Time Saved (hours/day × 25 days × hourly cost):	ETB _____
Paper/Printing Saved:	ETB _____
Revenue Recovered (forgotten invoices):	ETB _____
Rush Purchase Avoidance:	ETB _____
TOTAL ESTIMATED MONTHLY BENEFIT:	ETB _____
vs. System Cost:	ETB _____
NET MONTHLY GAIN:	ETB _____

Demo Script: "Let's do a quick calculation together. If you save just 2 hours per day across your team, and recover just 2 forgotten invoices per month, you've already paid for the system."

Demo Talking Points by Audience

For the Owner/MD

- "Complete visibility into your business from your phone"
- "Know exactly who owes you money, every day"
- "Your business data is safe and organized forever"
- "Professional image attracts bigger clients"

For the Accountant

- "No more month-end scramble - books are always up to date"
- "One-click payment reconciliation"
- "Automatic VAT calculation on every invoice"
- "Clear audit trail for every transaction"

For the Sales Team

- "Create quotations in 2 minutes, not 20"
- "Know instantly if you have stock before promising delivery"
- "Track your leads - never forget to follow up"
- "Look professional with branded documents"

For Production/Operations

- "Clear work orders - no more verbal instructions"
- "Know exactly what raw materials are needed"
- "Delivery notes ready when the job is done"
- "No confusion about what's been delivered"

ET Ethiopia-Specific Compliance Benefits

Compliance Area	ERP Support
VAT (15%)	Auto-calculated on invoices, clear records for ERCA
Withholding Tax (2%)	Can be configured in tax templates
TIN Tracking	Customer TIN stored, appears on invoices
Receipt Numbering	Sequential, unique, compliant numbering
Financial Records	10-year retention ready, meets audit requirements

Demo Script: *"When ERCA comes for an audit, you hand them the laptop and say 'search anything.' All your invoices, all your payments, all organized."*

Quick Reference: Demo Flow Summary

For a quick client demo, follow this condensed flow:

1. **Create Lead** → Show inquiry capture
2. **Convert to Customer** → Add address/contact
3. **Create Quotation** → Add item with specs
4. **Submit Quotation** → Show status change
5. **Create Sales Order** → Show artwork check
6. (Optional) **Create Work Order** → Show manufacturing
7. **Create Delivery Note** → Print gate pass
8. **Create Invoice** → Show outstanding
9. **Create Payment** → Show paid status
10. **Show Dashboard** → Summarize the completed cycle

Estimated Demo Time: 30-45 minutes

This testing guide ensures complete validation of Pana ERP v3.0 before client deployment.